

1/

```
stateDiagram-v2
    direction LR

    state "ECOSYSTEM SETUP" as Setup {
        MF1_Client_Onboarding : MF-1 Client Reg (incl. MF-11 TopUp, MF-13 Sub)
        MF2_Expert_Onboarding : MF-2 Expert Reg (incl. MF-11 TopUp, MF-13 Sub)
        MF3_TechTeam_Handoff : MF-3 Tech Team Handoff
        MF15_Dual_Role : MF-15 Dual-Role Switch
    }

    state "PATH A: AI PROJECT FLOW" as PathA {
        MF4_Elicitation : MF-4 Elicitation (RQ2 Core)
        MF5_Matching : MF-5 Matching (RQ1 Core)
        MF6_Bidding : MF-6 Bid & Connection
        MF7_Escrow : MF-7 Milestones & Escrow (RQ3 Core)
    }

    state "PATH B: MARKETPLACE" as PathB {
        MF9_Service_Publish : MF-9 Service Publish
        MF10_Service_Buy : MF-10 Service / Tech Discovery Buy
    }

    state "TRUST & INTEGRITY" as Trust {
        MF8_Disputes : MF-8 Dispute Resolution (RQ3 Core)
        MF12-Withdrawal : MF-12 Auto-Withdrawal
        MF14_Scope_Evo : MF-14 Add-On Protocol
        MF16_Verification : MF-16 Tier 4 Auto-Upgrade
        MF17_Admin_Monitor : MF-17 Admin Monitor
        MF18_Fraud_Detection : MF-18 Fraud Detection
    }

    [*] --> Setup
    Setup --> PathA : CEO initiates project
    Setup --> PathB : CEO browses services

    PathA --> Trust : Engagement Active
    PathB --> Trust : Payment Held

    Trust --> [*] : Funds Released / Disputed
```

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```
stateDiagram-v2
    direction TB

    state "Layer 1: Identity & Qualification" as L1 {
```

```

MF1_Client_Onboarding : MF-1 Client Reg & Sub
MF2_Expert_Onboarding : MF-2 Expert Reg & Verify
MF3_Tech_Handoff : MF-3 Tech Team Creation
MF13_Subscription : MF-13 Sub Purchase
MF15_Dual_Role : MF-15 Dual-Role Switch
}

state "Layer 2: Supply & Demand Generation" as L2 {
  MF4_Elicitation : MF-4 AI Elicitation (Path A Demand)
  MF9_Service_Pub : MF-9 Service Publishing (Path B Supply)
}

state "Layer 3: Discovery & Contracting" as L3 {
  MF5_Matching : MF-5 AI Matching
  MF6_Bidding : MF-6 Bid & Negotiation
}

state "Layer 4: Value Exchange & Delivery" as L4 {
  MF7_Milestones : MF-7 Milestones & Escrow
  MF14_AddOn : MF-14 Add-On Protocol
  MF10_Service_Buy : MF-10 Service Purchase (Path B)
}

state "Layer 5: Settlement, Trust & Infrastructure" as L5 {
  MF8_Disputes : MF-8 Dispute Resolution
  MF11_Wallet : MF-11 Wallet Top-Up
  MF12_Withdraw : MF-12 Auto-Withdrawal
  MF16_Tier4 : MF-16 Auto-Upgrade
  MF17_Admin : MF-17 Integrity Monitor
  MF18_Fraud : MF-18 Fraud Detection
}

[*] --> L1
L1 --> L2 : Define needs/supply
L2 --> L3 : Find counterparties
L3 --> L4 : Execute contract
L4 --> L5 : Settle & Evaluate

L5 --> L1 : Upgrade Tier / Switch Role
L5 --> L2 : Re-verify capabilities

```

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```

flowchart TD
  %% Styling
  classDef highwayA fill:#e1f5fe,stroke:#039be5,stroke-width:2px;
  classDef highwayB fill:#fff3e0,stroke:#fb8c00,stroke-width:2px;
  classDef infra fill:#f3e5f5,stroke:#8e24aa,stroke-width:2px;
  classDef layer fill:#fafafa,stroke:#bdbdbd,stroke-width:1px,stroke-
dasharray: 5 5;

```

```

%% Layers
subgraph L1 [Layer 1: Identity & Qualification]
    direction LR
    MF1[MF-1: Client Reg & Sub]:::highwayA
    MF2[MF-2: Expert Reg & Verify]:::highwayB
    MF3[MF-3: Tech Team Handoff]:::highwayA
    MF13[MF-13: Sub Purchase]:::infra
    MF15[MF-15: Dual-Role Switch]:::infra
end

subgraph L2 [Layer 2: Supply & Demand Generation]
    direction LR
    MF4[MF-4: AI Elicitation Engine]:::highwayA
    MF9[MF-9: Service Publishing]:::highwayB
end

subgraph L3 [Layer 3: Discovery & Contracting]
    direction LR
    MF5[MF-5: AI Matching]:::highwayA
    MF6[MF-6: Bid & Negotiation]:::highwayA
end

subgraph L4 [Layer 4: Value Exchange & Delivery]
    direction LR
    MF7[MF-7: Milestones & Escrow]:::highwayA
    MF14[MF-14: Add-On Protocol]:::highwayA
    MF10[MF-10: Service Purchase]:::highwayB
end

subgraph L5 [Layer 5: Settlement, Trust & Infrastructure]
    direction LR
    MF8[MF-8: Dispute Resolution]:::infra
    MF11[MF-11: Wallet Top-Up]:::infra
    MF12[MF-12: Auto-Withdrawal]:::infra
    MF16[MF-16: Tier 4 Auto-Upgrade]:::infra
    MF17[MF-17: Admin Monitor]:::infra
    MF18[MF-18: Fraud Detection]:::infra
end

%% Highway A (Path A: Project-Based)
MF1 -->|Needs AI| MF13
MF13 -->|Unlocks| MF4
MF4 -->|Hard-blocks CEO| MF3
MF3 -->|Returns truth| MF4
MF4 -->|Publishes Footprint| MF5
MF5 -->|Shortlist| MF6
MF6 -->|Connected| MF7
MF7 -->|Scope Evolves| MF14
MF14 -->|Appended| MF7

%% Highway B (Path B: Marketplace)
MF2 -->|Needs Pro| MF13
MF2 -->|Creates listing| MF9

```

```

MF9 -->|Visible to| MF1
MF1 -->|Buys| MF10

%% Layer 5 Interactions (Cross-cutting)
MF7 -->|Funds needed| MF11
MF7 -->|Approved| MF12
MF7 -->|Disputed| MF8
MF10 -->|Paid| MF12
MF2 -->|Links Bank| MF18
MF12 -->|Closed| MF16
MF16 -->|Upgrades| MF5
MF8 -->|Audited by| MF17
MF1 <-->|Switches| MF15

%% Cross-Layer routing
L1 -.-> L2
L2 -.-> L3
L3 -.-> L4
L4 -.-> L5
L5 -.-> L1

```

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```

stateDiagram-v2
    direction TB

    state "Layer 1: Identity & Qualification" as L1_Setup {
        direction LR
        state "MF-1 Client Reg" as MF1_Client
        state "MF-2 Expert Reg & Verify" as MF2_Expert
        state "MF-3 Tech Team Handoff" as MF3_TechTeam
        state "MF-11 Wallet Top-Up" as MF11_Wallet
        state "MF-13 Subscription Purchase" as MF13_Sub
        state "MF-15 Dual-Role Switch" as MF15_Dual
    }

    state "Layer 2: Supply & Demand Generation" as L2_Gen {
        direction LR
        state "MF-4 AI Elicitation (Path A)" as MF4_Elicitation
        state "MF-9 Service Publishing (Path B)" as MF9_Service
    }

    state "Layer 3: Discovery & Contracting" as L3_Contract {
        direction LR
        state "MF-5 AI Matching" as MF5_Matching
        state "MF-6 Bid & Negotiation" as MF6_Bidding
    }

    state "Layer 4: Value Exchange & Delivery" as L4_Delivery {
        direction LR
        state "MF-7 Milestones & Escrow" as MF7_Escrow
    }

```

```

state "MF-14 Add-On Protocol" as MF14_AddOn
state "MF-10 Service Purchase (Path B)" as MF10_BuyService
}

state "Layer 5: Settlement, Trust & Infrastructure" as L5_Trust {
    direction LR
    state "MF-8 Dispute Resolution" as MF8_Dispute
    state "MF-12 Auto-Withdrawal" as MF12_Withdraw
    state "MF-16 Tier 4 Auto-Upgrade" as MF16_Upgrade
    state "MF-17 Admin Monitor" as MF17_Admin
    state "MF-18 Fraud Detection" as MF18_Fraud
}

[*] --> L1_Setup : User enters ecosystem

L1_Setup --> L2_Gen : Identity established and Pro unlocked

%% HIGHWAY A - Project-Based Flow (Traces through Layers 2 -> 3 -> 4)
MF4_Elicitation --> MF5_Matching : HIGHWAY A - Spec PUBLISHED
MF5_Matching --> MF6_Bidding : HIGHWAY A - Shortlist generated
MF6_Bidding --> MF7_Escrow : HIGHWAY A - Expert SELECTED and Connected
MF7_Escrow --> MF14_AddOn : HIGHWAY A - Sprint hits SCOPE EVOLUTION
MF14_AddOn --> MF7_Escrow : Add-on appended

%% HIGHWAY B - Marketplace Flow (Bypasses Layer 3, goes 2 -> 4)
MF9_Service --> MF10_BuyService : HIGHWAY B - Client clicks Buy Now

%% Layer Transitions
L4_Delivery --> L5_Trust : Milestone APPROVED or DISPUTED

L5_Trust --> L1_Setup : MF-16 upgrades profile or MF-15 switches context
L5_Trust --> [*] : Ledger settled and funds withdrawn

note right of L5_Trust
    MF-17 and MF-18 run continuously
    as cross-cutting integrity guards
    across all layers.
end note

```

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Part 1: High-Level Navigation Map

The AltTasker UI is **role-driven**. The navigation bar completely re-renders based on the JWT's **active_role**.

```

stateDiagram-v2
    direction TB

    state "Authentication Stack" as Auth {

```

```

    Login_Register : Login / Register
    Role_Switcher : Role Switcher (Top Nav)
}

state "CLIENT_CEO View" as CEO {
    CEO_Dash : CEO Dashboard
    Elicitation : Elicitation Wizard (MF-4)
    Shortlist : Shortlist View (MF-5)
    Bid_Review_CEO : Bid Review (Business/Surface C&D)
    Project_Mgmt_CEO : Project Management (Fund / Approve Business)
    Marketplace : Marketplace Browser (MF-10)
}

state "CLIENT_TECH_TEAM View" as Tech {
    Tech_Dash : Tech Dashboard
    Handoff : Architecture Handoff (MF-4 Stage 4)
    Bid_Review_Tech : Bid Review (Technical/Surface B)
    Sprint_View : Sprint View (Comment/Resolve Blockers)
    Signoff_Tech : Sign-off (Technical Milestones)
}

state "EXPERT View" as Expert {
    Expert_Dash : Expert Dashboard
    Profile_Verify : Profile & Verification (MF-2)
    Market_Browse : Project Market (View Artifact A)
    Bid_Workspace : Bid Workspace (MF-6)
    Delivery : Delivery Workspace (Sprint/DoD/Submit)
    Wallet_Withdraw : Wallet & Withdrawal (MF-11/12)
}

Auth --> CEO : active_role = CLIENT, subtype = CEO
Auth --> Tech : active_role = CLIENT, subtype = TECH_TEAM
Auth --> Expert : active_role = EXPERT

CEO --> Role_Switcher : If roles.length > 1
Tech --> Role_Switcher : (Disabled - Tech cannot switch)
Expert --> Role_Switcher : If roles.length > 1

Role_Switcher --> CEO
Role_Switcher --> Expert

```

Part 2: Screen Inventory & Components

1. CLIENT / CEO Screens

Goal: Help non-technical users define problems, pick the right expert, and manage budgets.

Screen Name	Core Purpose	Key UI Components	Mapped Flows
-------------	--------------	-------------------	--------------

Screen Name	Core Purpose	Key UI Components	Mapped Flows
CEO Dashboard	Command center	Wallet Balance Card, Pro Subscription Badge, "Start New AI Project" CTA, Active Project Cards	MF-1, MF-11, MF-13
AI Elicitation Wizard	Guided conversation	Step Indicator (1-5), Chat-style Input, Archetype Selector Cards, "Inject Phase 0" system message, Hard-Block Modal (Generate Handoff Link)	MF-4, MF-3
Shortlist View	Compare matched experts	Match Cards (Strong/Qualified labels), Seam Gap Map Visual , Stack Tag Overlaps, "Select Expert" button	MF-5
Bid Review (CEO)	Negotiate business terms	Bid Version History, Tech Team Recommendation Badge, Surface C (Price Negotiation Panel), Surface D (Override Modal with mandatory reason input)	MF-6
Project Management	Track budget & milestones	Milestone Timeline, "Fund Milestone" button (generates VietQR modal), Business Milestone Sign-off button	MF-7
Marketplace Browser	Buy direct services	Service Cards, "Buy Now" button, Service Detail Page	MF-9, MF-10

2. CLIENT / TECH_TEAM Screens

Goal: Provide technical reality, validate expert approaches, and track code-level delivery.

Screen Name	Core Purpose	Key UI Components	Mapped Flows
Tech Dashboard	Technical tasks	Pending Handoff Alerts, Active Sprint Alerts, Technical Sign-off Queue	MF-3, MF-7
Architecture Handoff	Populate Artifact B	Stack Tag Multi-select, Integration Method Dropdown, File Upload Zone (schema/topology)	MF-4 (Stage 4)
Bid Review (Tech)	Validate tech approach	Bid Version History, Surface B (Request Revision Modal - targeting specific component) , Recommend/Not Recommend Toggle	MF-6
Sprint View	Monitor delivery	Sprint Status Badge (On Track/Blocker), Blocker Resolution Input , Sprint Comment Thread	MF-7, MF-14
Sign-off (Tech)	Approve technical deliverables	DoD Checklist Read-only View, Technical Acceptance Criteria, Sign-off Button	MF-7

3. EXPERT Screens

Goal: Prove capability, win bids, deliver work, get paid automatically.

Screen Name	Core Purpose	Key UI Components	Mapped Flows
Expert Dashboard	Earnings & tasks	Earnings Chart, Pro Badge, Bank Link Status, Active Bids/Sprints, "Claim Seams" CTA	MF-2, MF-12
Profile & Verification	Build taxonomy map	Domain Depth Selector, Seam Cards (Tier 1/2/3 badges) , "Verify with AI" button (Tier 2 file upload / Tier 3 scenario chat)	MF-2
Project Market	Find work	Artifact A Viewer, Surface A (Spec Clarification Panel - Ask question) , "Submit Bid" button	MF-5, MF-6
Bid Workspace	Construct & negotiate bid	3-Component Form (Tech/Delivery/Finance), Surface B (Revision Update Form - only flagged component editable) , Surface C (Accept/Counter Price)	MF-6
Delivery Workspace	Execute work	Milestone Timeline, Sprint Plan Editor, DoD Checklist (Checkboxes with Completion Notes) , "Submit Deliverable" button	MF-7
Wallet & Withdrawal	Manage funds	Top-Up VietQR Display, Withdrawal Form (Amount input), Withdrawal History (Pending/Completed)	MF-11, MF-12

Part 3: UI/UX State-Driven Rules

This is the most important section for your frontend teammate. AI Tasker's UI is heavily dictated by backend state machines. Components must conditionally render based on these rules:

Rule 1: The Subscription Guard (Feature Gating)

- **IF** `user.subscription_tier === 'free'`:
 - "Start New AI Project" button is disabled/hidden.
 - Clicking AI features triggers a `403` interceptor → Show **"Upgrade to Pro" Upsell Modal**.
 - Expert Seam Verification buttons (Tier 2/3) are locked.

Rule 2: The Elicitation Hard Block (MF-4 Stage 4)

- **IF** `project.self_technical === false` AND `infra_signals === 'HIGH'`:
 - The Elicitation Wizard *must not* allow the CEO to proceed to Stage 4.
 - Show a full-screen "Hard Block" overlay: "Your project involves live infrastructure..." with a "Generate Handoff Link" button.

Rule 3: Milestone Submission Gate (DoD)

- **IF** `milestone.state === 'IN_PROGRESS'`:

- The "Submit Deliverable" button **must** be disabled UNTIL the frontend verifies that *all* DoD items with `is_required = true` are checked `COMPLETED`.
- If a required item is marked `NOT_APPLICABLE` by the expert, the UI should throw a validation error (backend will throw 403).

Rule 4: State-Gated Artifact Access

- **Artifact A:** Visible to all matched experts and the client.
- **Artifact B:**
 - **IF** `engagement.state < CONNECTED`: UI must *not* render the Artifact B tab/endpoint for the Expert.
 - **IF** `engagement.state >= CONNECTED`: Artifact B tab appears for Expert and Tech Team.

Rule 5: Bid Surface Routing (Non-Linear Bids)

- The Bid Detail screen must route the user based on the `bid.state`:
 - `TECH_REVIEW`: Tech Team sees the review form.
 - `REVISION_REQUESTED`: Expert sees *only* the flagged component text area editable; other fields are read-only.
 - `CONFLICT_PENDING`: CEO sees the Override Modal (must input reason before viewing bid).

Rule 6: The Dual-Role Switcher

- **IF** `user.roles.length > 1`:
 - Render a persistent toggle in the Top Navigation (e.g., "Switch to Expert").
 - Clicking it hits `/auth/switch-role`, returns a new JWT, and the entire app shell (nav, dashboard) re-mounts.
 - **Self-Exclusion:** Even when in Expert mode, the Project Market shortlist API will automatically filter out any project where the user is the CEO. No special UI needed, but good to know.

Rule 7: The Escrow & VietQR Flow

- When a CEO clicks "Fund Milestone":
 - Show a Modal with a loading spinner ("Generating secure QR...").
 - Render the VietQR image and a 24-hour countdown timer.
 - Use WebSockets or polling: Once the SePay IPN hits the backend, the modal should auto-close and the Milestone state should live-update to `FUNDED` without a manual page refresh.

Rule 8: Sprint Status & Add-On Protocol

- **IF** Expert selects `Sprint Status = BLOCKER` and `blocker_type = SCOPE_EVOLUTION`:
 - The UI must immediately prompt: *"This requires a scope change. Fill out the Add-On Proposal form."*
 - This links the Sprint UI directly to the MF-14 Add-On creation flow.